RETAIL SALES & SUPPLY CHAIN ANALYTICS DASHBOARD

Integrated Technical and Business Analysis Report

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1. Project purpose and business problem

The objective of this project was to convert a multi-year retail dataset into an interactive management reporting solution. The source data contained sales transactions, products, customers, locations, shipping information and returns. In its raw form, the data could not quickly answer important questions such as: Which categories create profit rather than only revenue? Which region and customer segment produce the strongest performance? Where are delivery delays and returns concentrated? The final Power BI solution was therefore designed to combine technical data preparation with practical business analysis.

2. Data preparation, modelling and calculations

The dataset was imported into Power BI and prepared in Power Query. The preparation stage included reviewing column names and data types, checking for missing or inconsistent values, removing fields that were not required for analysis, and creating date attributes for Year, Quarter and Month reporting. Month names were sorted by month number to prevent alphabetical sorting, and the geographic and categorical fields were checked so they could be used consistently in maps, slicers and charts.

A structured model was then created by relating the main sales/order data to supporting customer, product, return and date information. This allowed measures to respond dynamically when users filtered by region, year, category, customer segment or shipping method. Measures were preferred over static calculated columns for key KPIs because measures recalculate within the current filter context.

Example DAX measure	Analytical purpose
Total Sales = SUM(Sales[Sales])	Measures revenue across the selected period, region or category.
Total Profit = SUM(Sales[Profit])	Separates profitable performance from high sales volume.
Profit Margin = DIVIDE([Total Profit], [Total Sales])	Shows profit earned for each dollar of sales.
Average Order Value = DIVIDE([Total Sales], [Total Orders])	Measures the typical financial value of an order.
Return Rate = DIVIDE([Returned Orders], [Total Orders])	Monitors the proportion of orders that were returned.
Average Delivery Days	Evaluates logistics speed by ship mode and region.
On-Time Delivery %	Measures the share of deliveries completed within the defined target.

3. Executive performance overview

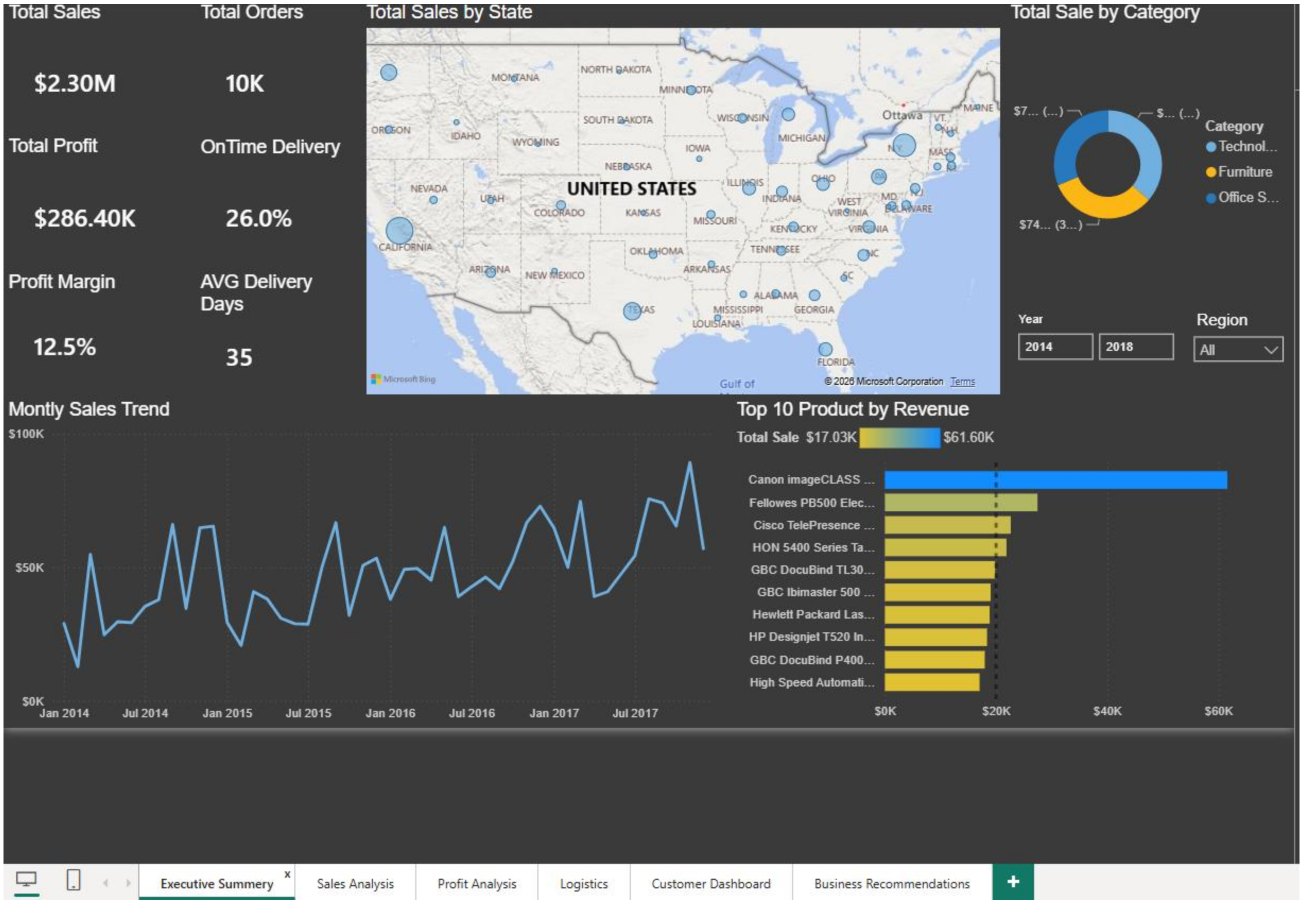

Figure 1. Executive Summary dashboard

The executive dashboard establishes the overall performance baseline. Total sales reached approximately $2.30 million from about 10,000 order records, producing $286.40 thousand in profit and a 12.5% profit margin. These results show that the business was profitable overall, but the logistics KPIs reveal an important weakness: average delivery time was 35 days and the on-time delivery KPI was only 26%. Financial performance was therefore stronger than operational delivery performance.

The monthly sales trend from 2014 to 2017 fluctuated but generally strengthened toward the later reporting periods, with several peaks above $60,000 and the strongest months approaching $90,000. The state map shows that sales were geographically concentrated in major markets, particularly California and several eastern states. The product ranking also revealed concentration risk: the Canon image CLASS product generated about $61.60 thousand in revenue, substantially higher than the next products. This means a small number of products contributed a disproportionate share of sales and should be monitored closely for inventory availability and supplier continuity.

Sales and customer interpretation

The detailed sales and customer pages expanded this overview. The Consumer segment generated $1,161,401 in sales and represented the strongest customer segment. The West was the highest-sales region at $725,458. These findings suggest that the business should protect its position in the West while continuing to target Consumer customers through retention, loyalty and category-specific campaigns. The customer dashboard also made it possible to identify leading customers and states, helping management move from broad market reporting to targeted account and geographic action.

4. Profitability analysis

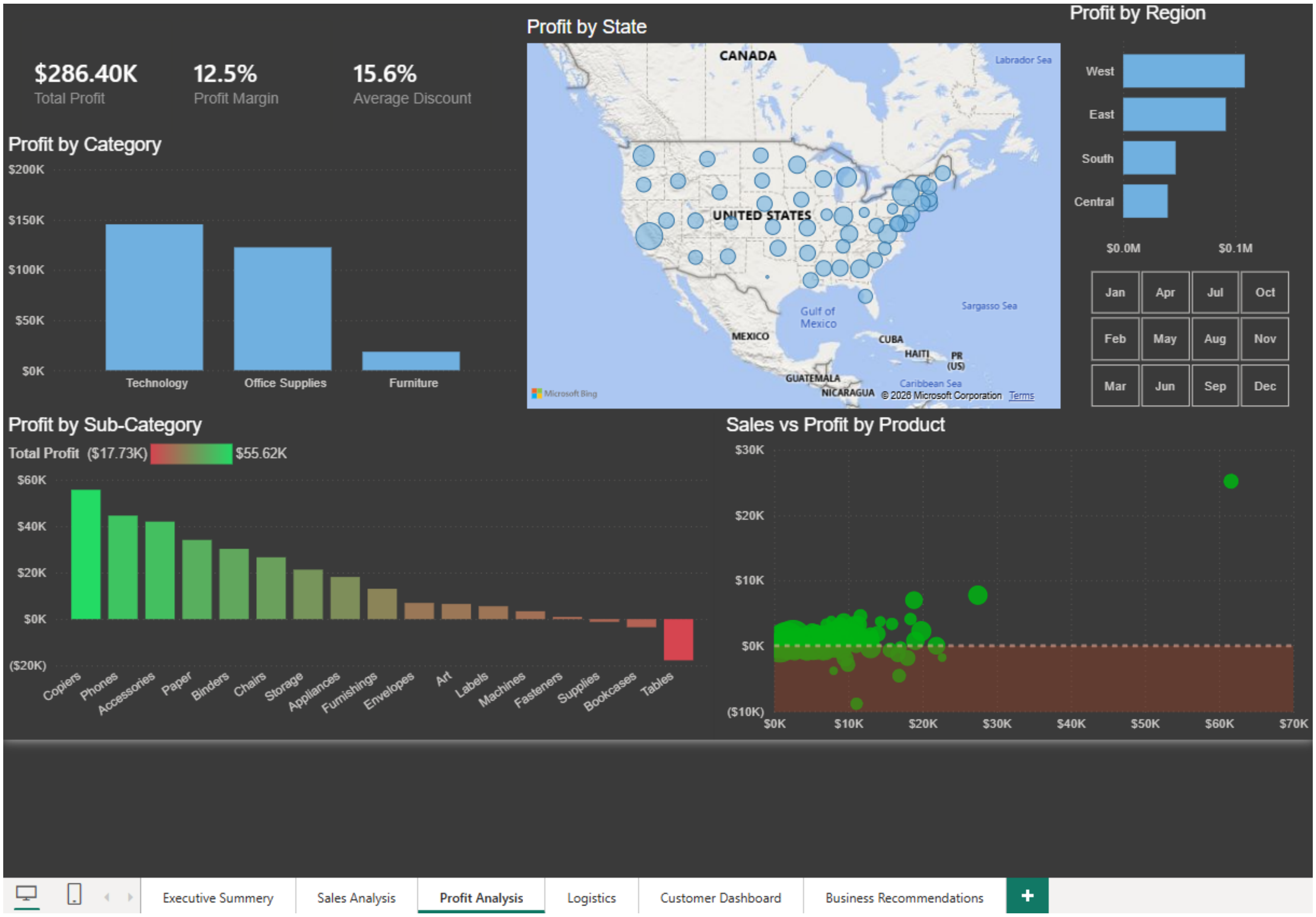

Figure 2. Profit Analysis dashboard

Revenue alone did not explain business quality, so the profitability page compared profit by category, sub-category, state, region and product. Technology was the most profitable category, generating approximately $145,455 in profit. Office Supplies also produced a strong positive result, while Furniture contributed far less profit. At sub-category level, Copiers were the strongest contributor at roughly $55.6 thousand, followed by Phones and Accessories. In contrast, Tables recorded a loss of approximately $17,725 and were the lowest-profit sub-category.

The sales-versus-profit scatter plot was particularly useful because it distinguished high-revenue products from genuinely profitable products. A product positioned far to the right but close to, or below, the zero-profit line may generate sales without creating sufficient value. The chart also showed a small number of profitable outliers, including one product with sales above $60,000 and profits near $25,000. Management should therefore avoid judging products only by revenue and should use profit contribution and margin when making pricing, promotion and stocking decisions.

Regional profit followed a similar pattern to sales: the West was strongest, followed by the East, while Central produced the lowest regional profit. This does not mean Central should automatically be reduced, but it does justify investigation into product mix, discounting, shipping cost and customer composition in that region. The 15.6% average discount is another important factor because discounting can increase revenue while weakening margin. Discount levels should be evaluated against product profitability rather than applied uniformly.

5. Logistics and return analysis

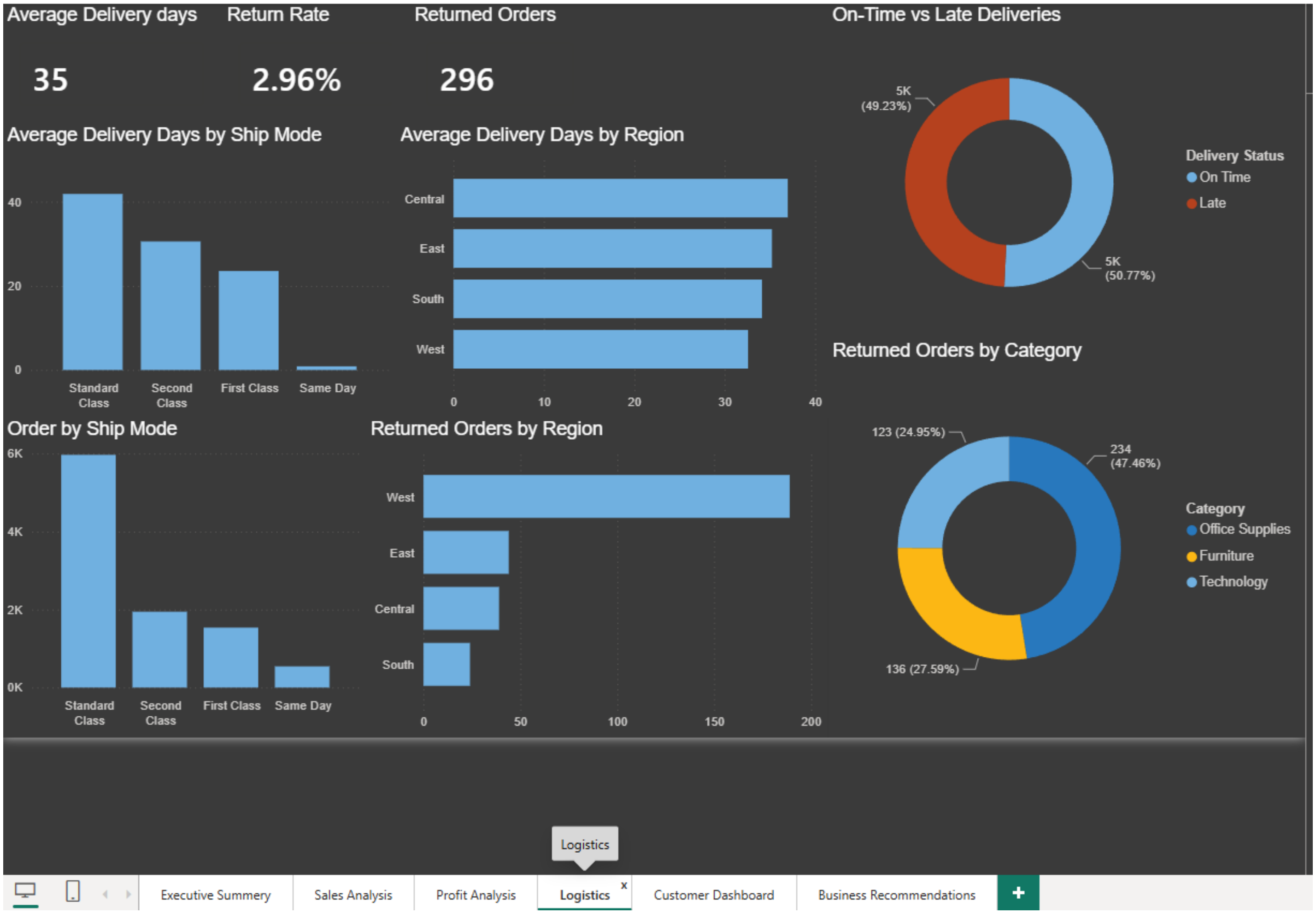

Figure 3. Logistics dashboard

The logistics dashboard identified the clearest operational improvement opportunity. Average delivery time was 35 days, with Standard Class taking approximately 42 days, Second Class being about 31 days and First Class being around 24 days. Same Day shipping was much faster, but it represented only a small share of total volume. Standard Class handled approximately 6,000 order records, so even a modest improvement in this shipping method could affect a large proportion of customers.

Regional delivery performance also varied. Central had the longest average delivery time at roughly 37 days, while West was the fastest at around 33 days. This supports a targeted logistics review rather than a single business-wide response. Carrier selection, warehouse allocation, order processing time and regional shipping routes should be examined first in Central.

There were 296 returned orders, equal to a 2.96% return rate. The West recorded the largest number of returns, although this should be interpreted alongside its higher sales volume. By category, Office Supplies generated 234 returned order records (47.46%), Furniture 136 (27.59%) and Technology 123 (24.95%). The Office Supplies result is large enough to justify product-level investigation into quality issues, inaccurate descriptions, damage in transit or customer expectation gaps. A low overall return rate is positive, but the category concentration shows that improvement can still be targeted.

Operational implication

Improving delivery performance is likely to protect customer satisfaction without requiring the business to change its strongest sales strategy. Because Consumer customers and the West already perform strongly, the priority is to improve the service experience around those sales while addressing slower delivery in Central and the return concentration in Office Supplies.

6. Integrated findings, recommendations and conclusion

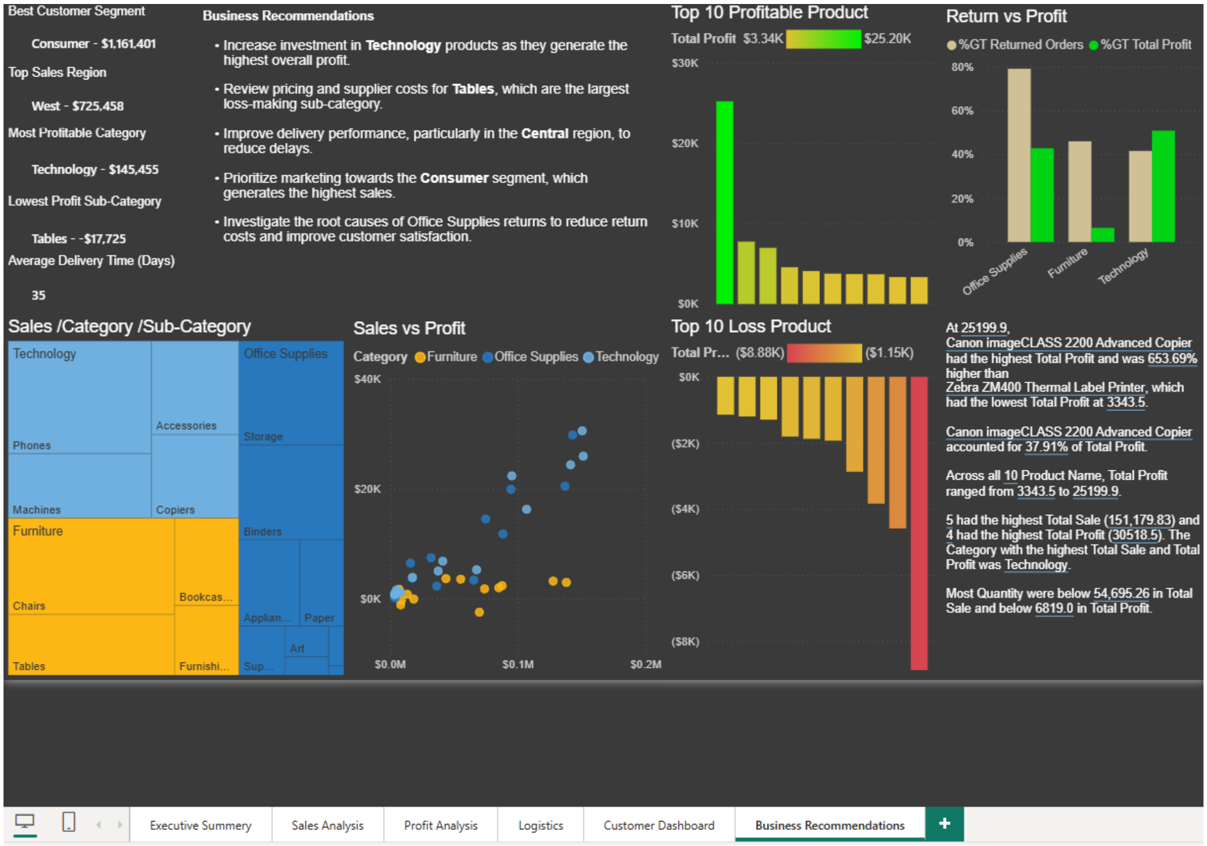

Figure 4. Business Recommendations dashboard

Increase investment in Technology: Technology produced the highest category profit ($145,455). Inventory planning, product availability and marketing should priorities high-margin Technology lines, while avoiding overreliance on a single top product.

Correct the Tables profitability problem: Tables generated the largest sub-category loss (-$17,725). Management should review purchase cost, freight, discounting, returns and selling price before increasing volume in this area.

Improve Central-Region delivery performance: Central recorded the longest average delivery time. A focused review of warehouse routing, carrier performance and order processing can improve service more efficiently than a broad, untargeted intervention.

Priorities the Consumer segment: Consumer customers generated $1.161 million in sales, making them the largest segment. Retention campaigns, personalized offers and cross-selling should be concentrated where expected value is highest.

Investigate Office Supplies returns: Office Supplies represented 47.46% of returned order records. Product-level root-cause analysis should identify whether the issue is quality, packaging, delivery damage or incorrect customer expectations.

Use profit and service KPIs together: The business is profitable, but the 35-day average delivery time and low on-time KPI show that revenue growth should not be assessed separately from customer service and operational efficiency.

Conclusion

This project combined Power Query, data modelling, DAX and dashboard design with a clear business narrative. The analysis showed $2.30 million in sales and $286.40 thousand in profit, with Technology, Consumer customers and the West region driving the strongest results. It also identified loss-making Tables, slow Central-region delivery and concentrated Office Supplies returns. The final solution connects performance measures to practical management actions and demonstrates end-to-end Power BI and business analysis capability.